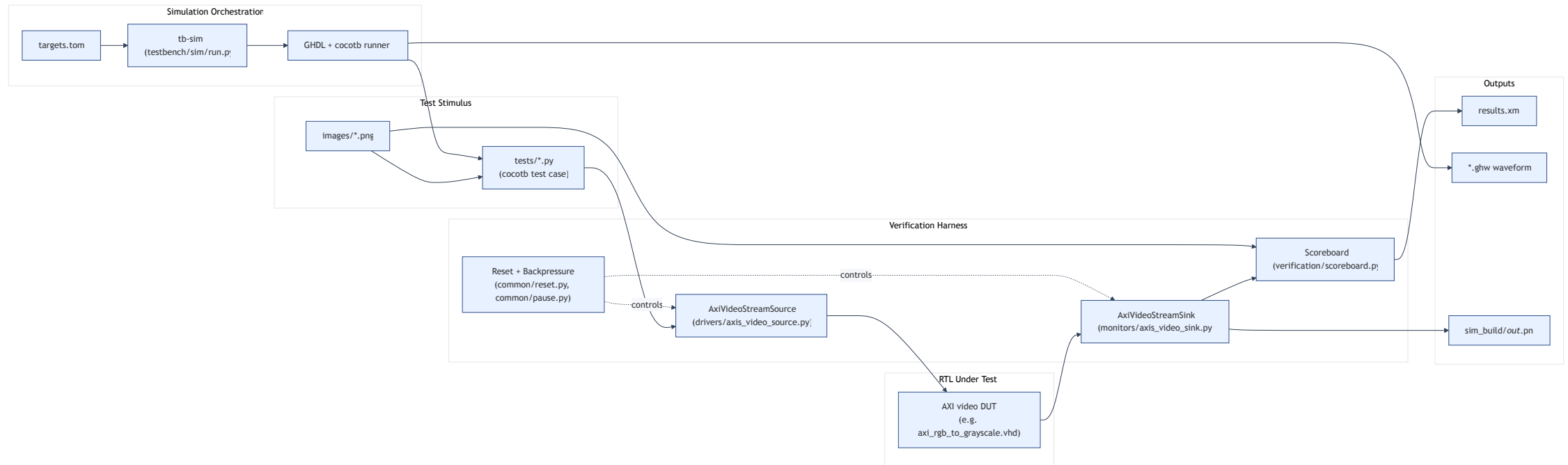


Realtime Image Processing

Project State (Verification + RGB2GRAY IP)

- Scope: testbench maturity, AXI4-Stream protocol confidence, and AXI_RGB2GRAY IP packaging status
- Repo: `realtime-image-processing`
- Date: 2026-02-13

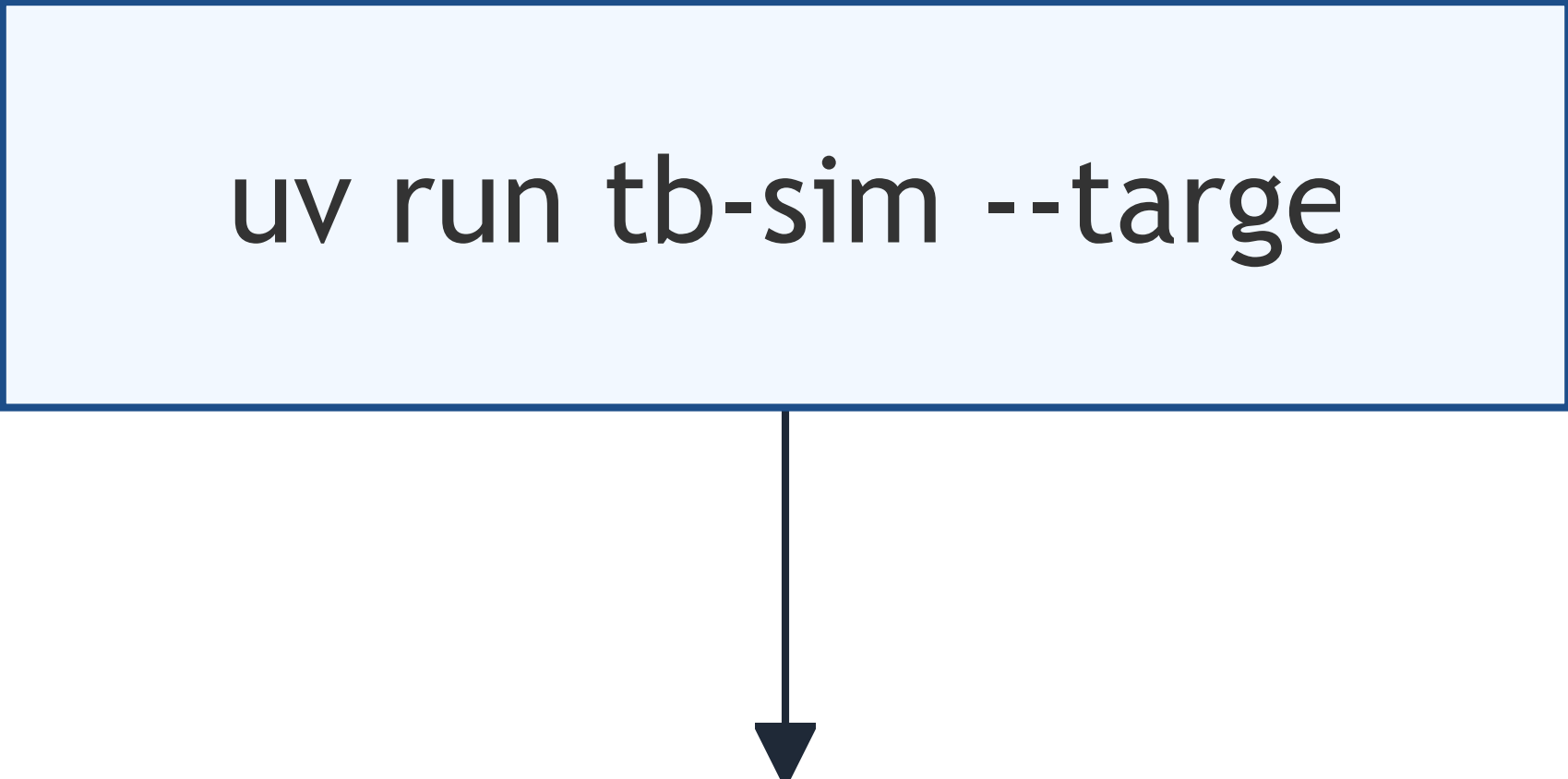
Verification Framework Architecture



- Cocotb test modules drive source/sink adapters and compare output via scoreboard.
- `tb-sim` resolves target configuration and builds DUT RTL through GHDL.

Verification Execution Flow

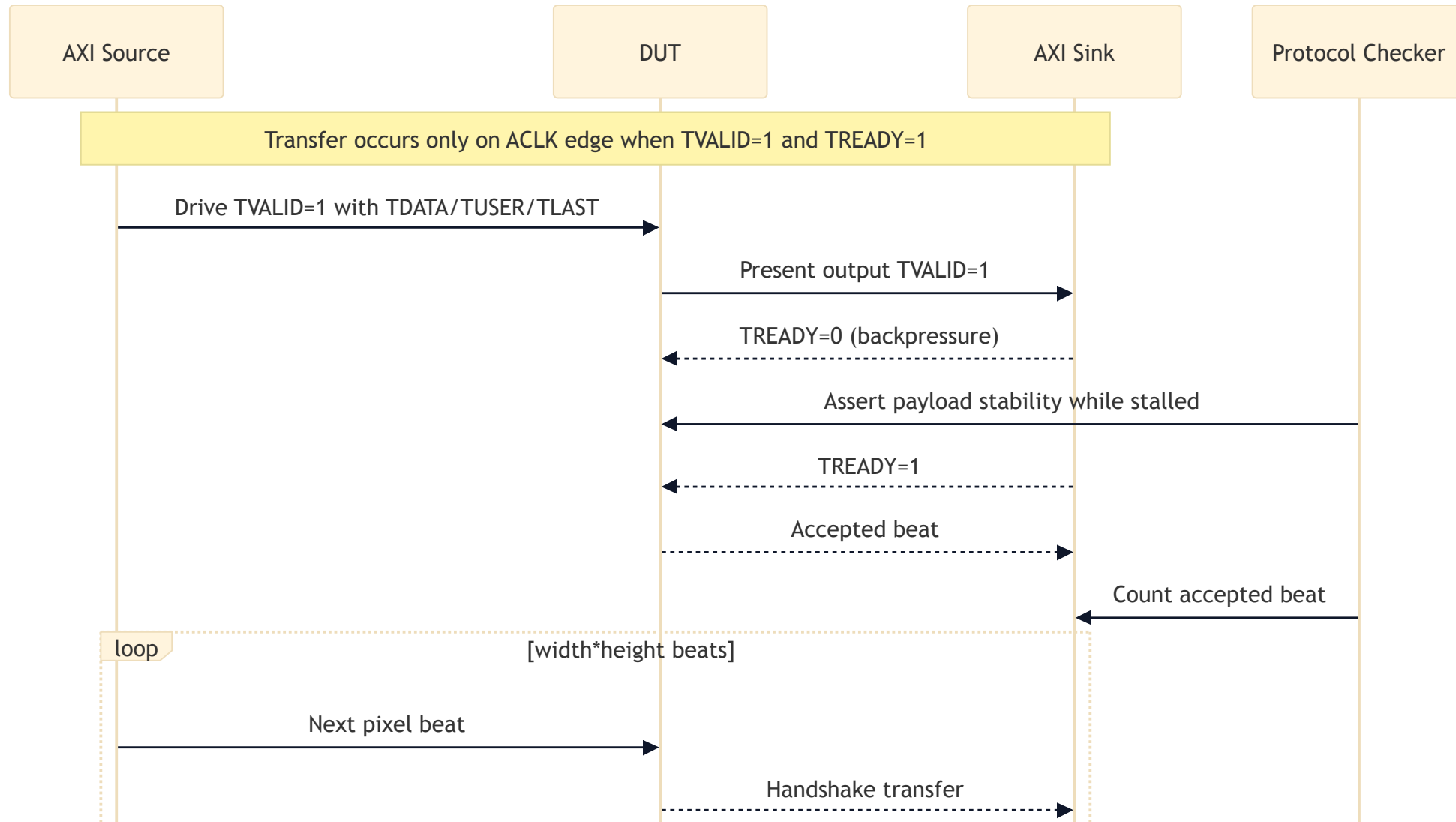
```
uv run tb-sim --target
```



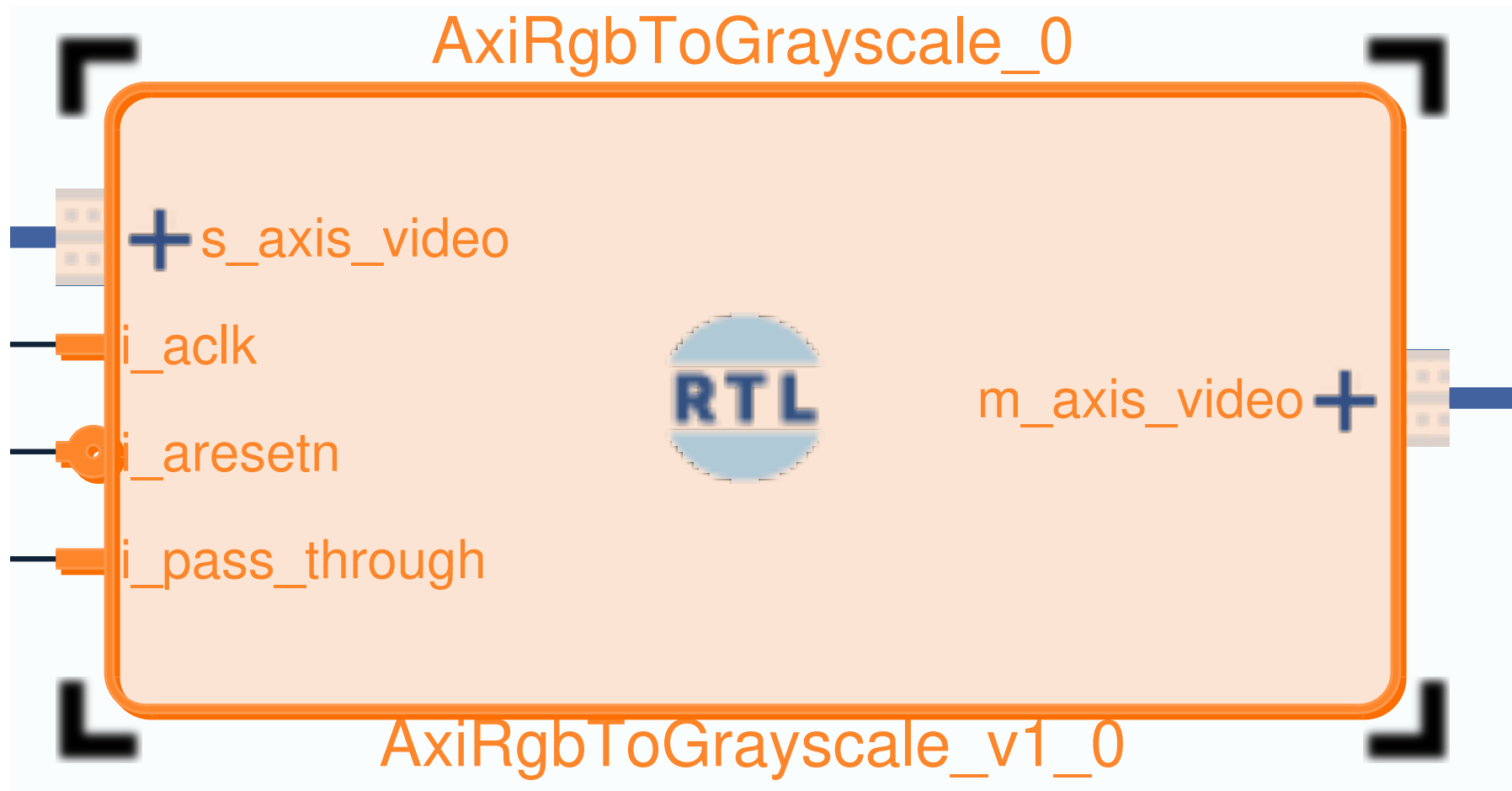
```
graph TD; A[uv run tb-sim --target] --> B[sim /run -v loads target]
```

```
sim /run -v loads target
```

AXI4-Stream Handshake Narrative

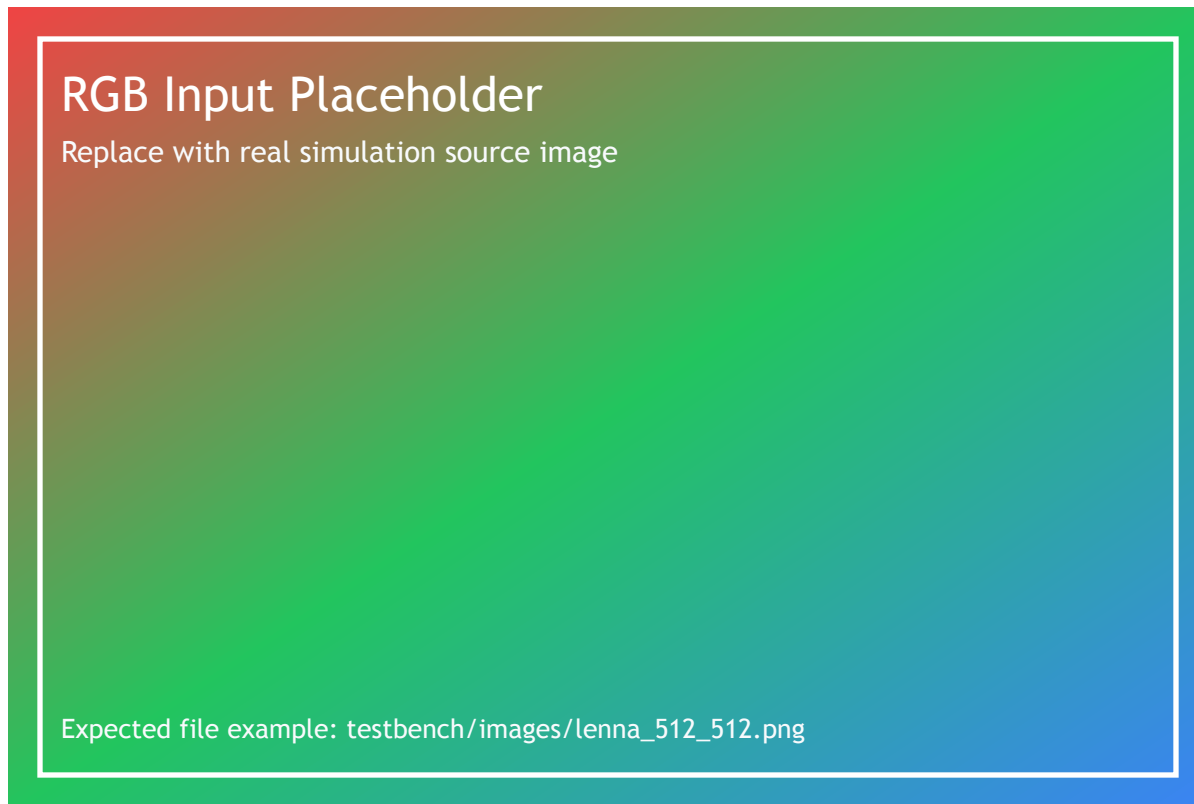


AXI_RGB2GRAY IP Core (Current Snapshot)

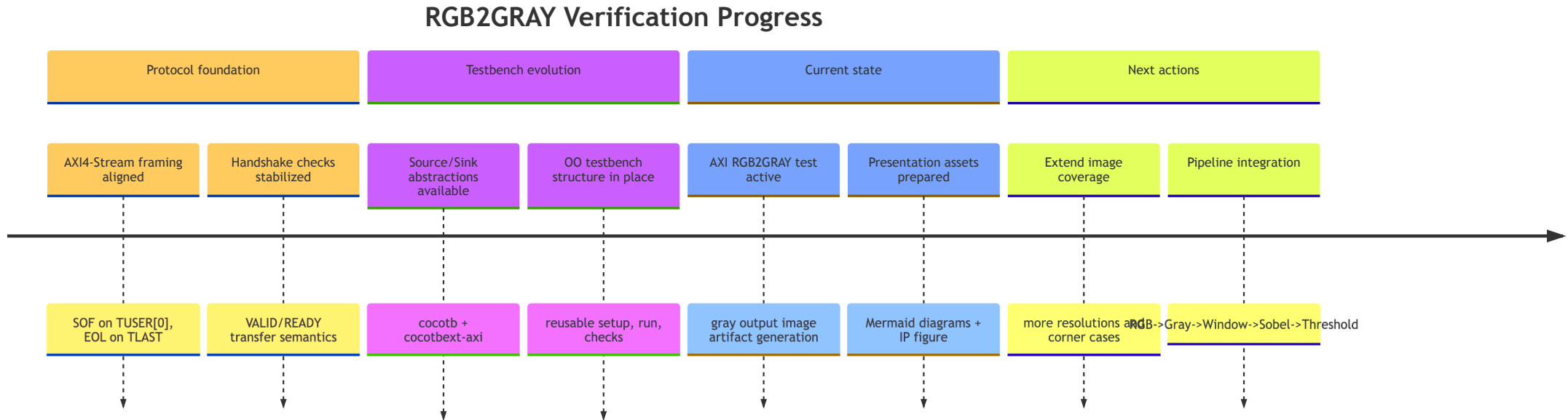


- Core HDL:

Simulation Evidence Placeholders



Current Status, Risks, Next Actions



- Open risks:
 - Protocol regressions across additional resolutions and pause patterns.
 - Integration mismatches when chaining into the full Sobel pipeline.
- Next actions: